

SQL Exercise - Advanced concepts

Exercise 1: Ranking and Window Functions

Solution :

```
CREATE DATABASE PracticeDB;
```

```
GO
```

```
USE PracticeDB;
```

```
GO
```

```
CREATE TABLE Products
```

```
(
```

```
    ProductID INT PRIMARY KEY,
```

```
    ProductName VARCHAR(50),
```

```
    Category VARCHAR(50),
```

```
    Price DECIMAL(10,2)
```

```
);
```

```
SELECT * FROM Products;
```

```
INSERT INTO Products VALUES
```

```
(1,'Laptop','Electronics',80000),
```

```
(2,'Mobile','Electronics',80000),
```

```
(3,'Tablet','Electronics',60000),
```

```
(4,'Smart Watch','Electronics',40000),
```

```
(5,'Sofa','Furniture',50000),
```

```
(6,'Dining Table','Furniture',50000),
```

```
(7,'Chair','Furniture',20000),
```

```
(8,'Cupboard','Furniture',15000),
(9,'Jacket','Clothing',6000),
(10,'Jeans','Clothing',6000),
(11,'Shirt','Clothing',4000),
(12,'T-Shirt','Clothing',2000);
```

Find Top 3 Products Using ROW_NUMBER() :

```
WITH ProductRanks AS (
    SELECT ProductName,
           Category,
           Price,
           ROW_NUMBER() OVER
           (
               PARTITION BY Category
               ORDER BY Price DESC
           ) AS RowNum
    FROM Products
) SELECT * FROM ProductRanks WHERE RowNum <= 3;
```

Output :

Results		Messages		
	ProductName	Category	Price	RowNum
1	Jacket	Clothing	6000.00	1
2	Jeans	Clothing	6000.00	2
3	Shirt	Clothing	4000.00	3
4	Laptop	Electronics	80000.00	1
5	Mobile	Electronics	80000.00	2
6	Tablet	Electronics	60000.00	3
7	Sofa	Furniture	50000.00	1
8	Dining Table	Furniture	50000.00	2
9	Chair	Furniture	20000.00	3

Top 3 Products Using RANK() :

```
WITH ProductRanks AS (  
    SELECT ProductName,  
           Category,  
           Price,  
           RANK() OVER  
           (  
               PARTITION BY Category  
               ORDER BY Price DESC  
           ) AS RankNo  
    FROM Products  
) SELECT * FROM ProductRanks WHERE RankNO <= 3;
```

Output :

Results		Messages		
	ProductName	Category	Price	RankNo
1	Jacket	Clothing	6000.00	1
2	Jeans	Clothing	6000.00	1
3	Shirt	Clothing	4000.00	3
4	Laptop	Electronics	80000.00	1
5	Mobile	Electronics	80000.00	1
6	Tablet	Electronics	60000.00	3
7	Sofa	Furniture	50000.00	1
8	Dining Table	Furniture	50000.00	1
9	Chair	Furniture	20000.00	3

Top 3 Products Using DENSE_RANK() :

```
WITH ProductRanks AS (  
    SELECT ProductName,  
           Category,  
           Price,  
           DENSE_RANK() OVER  
           (  
               PARTITION BY Category  
               ORDER BY Price DESC  
           ) AS DenseRankNo  
    FROM Products  
) SELECT * FROM ProductRanks WHERE DenseRankNO <= 3;
```

Output :

	ProductName	Category	Price	DenseRankNo
1	Jacket	Clothing	6000.00	1
2	Jeans	Clothing	6000.00	1
3	Shirt	Clothing	4000.00	2
4	T-Shirt	Clothing	2000.00	3
5	Laptop	Electronics	80000.00	1
6	Mobile	Electronics	80000.00	1
7	Tablet	Electronics	60000.00	2
8	Smart Watch	Electronics	40000.00	3
9	Sofa	Furniture	50000.00	1
10	Dining Table	Furniture	50000.00	1
11	Chair	Furniture	20000.00	2
12	Cupboard	Furniture	15000.00	3